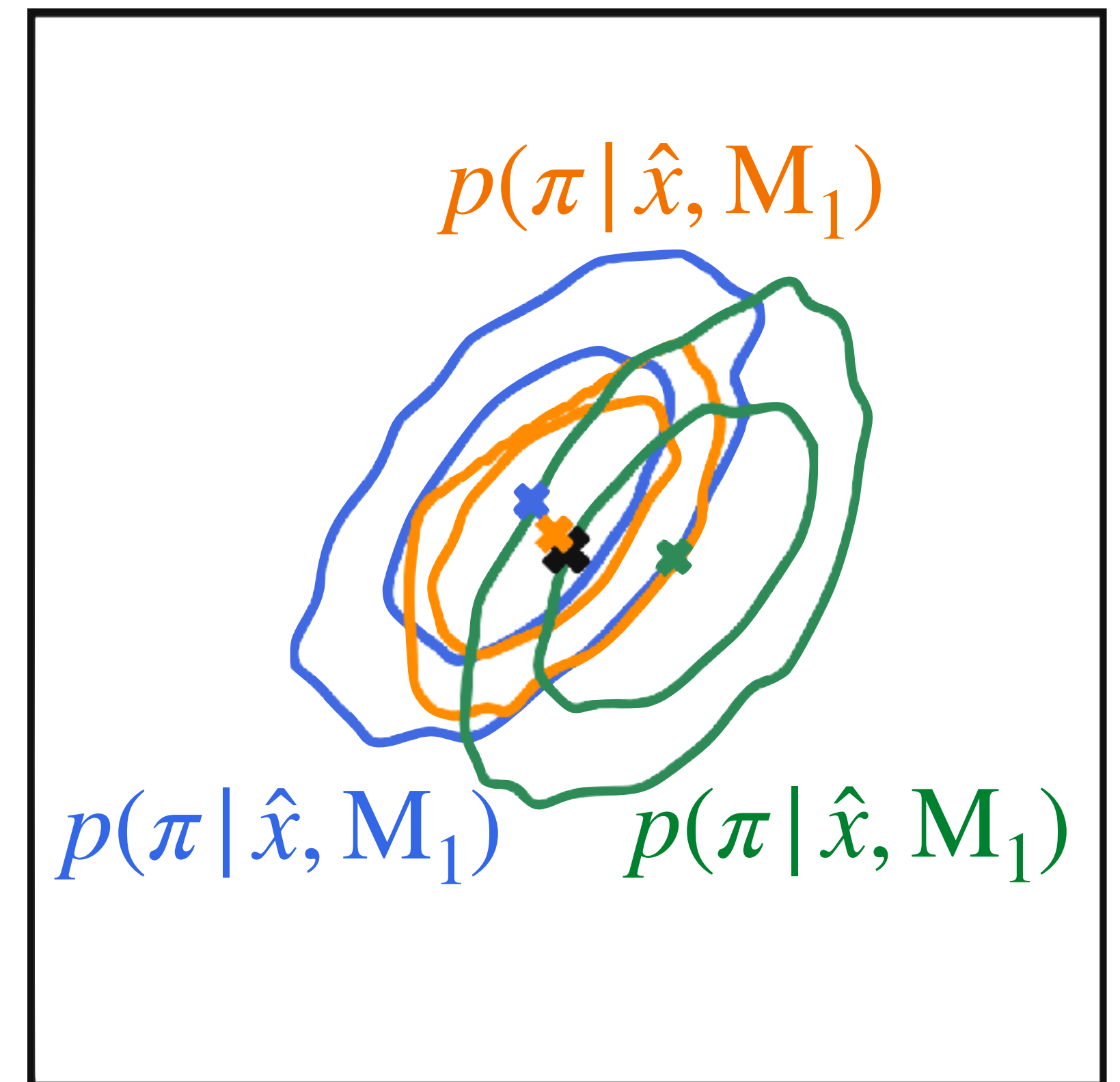


Bayesian analysis

Jed Homer LMU/ODSL



Bayesian Inference

- **Observational cosmology:** inferences about Universe by assigning a **posterior** PDF

$$p(\pi | \hat{x}) = \frac{p(\hat{x} | \pi)p(\pi)}{p(\hat{x})}$$

- Construct a PDF of physical parameters π from data $\hat{x}$

Bayesian Inference

- **Posterior PDF** is built from a **likelihood PDF** and a **prior PDF**

$$p(\pi | \hat{x}) = \frac{p(\hat{x} | \pi)p(\pi)}{p(\hat{x})}$$

- Usually we approximate the likelihood.
- The **evidence** $p(\hat{x}) = \int d\pi p(\hat{x} | \pi)p(\pi)$ is usually ignored for inferring π .

Likelihood PDF

- Inference requires a likelihood **PDF** which sets the limit of **information in $\hat{x}$ on π**

$$p(\hat{x} \mid x[\pi], \pi)$$

- Data $\hat{x}$ scatters around the expectation $x[\pi]$ as a function of π

Likelihood PDF

- Inference requires a likelihood **PDF** which sets the limit of **information in $\hat{x}$ on π**

$$p(\hat{x} \mid x[\pi], \pi)$$

e.g. $\mathcal{G}[\hat{x} \mid x[\pi], \Sigma]$

$$\propto \exp \left[-\frac{1}{2} (\hat{x} - x[\pi])^T \Sigma^{-1} (\hat{x} - x[\pi]) \right]$$

Gaussian PDF

$$\mathcal{G} [\hat{x} | x[\pi], \Sigma] = \frac{1}{|2\pi\Sigma|} \exp \left[-\frac{1}{2} (\hat{x} - x[\pi])^T \Sigma^{-1} (\hat{x} - x[\pi]) \right]$$

- A Gaussian PDF is characterised by μ and Σ .

Gaussian PDF

$$\mathcal{G} [\hat{x} | x[\pi], \Sigma] = \frac{1}{|2\pi\Sigma|} \exp \left[-\frac{1}{2} (\hat{x} - x[\pi])^T \Sigma^{-1} (\hat{x} - x[\pi]) \right]$$

- A Gaussian PDF is characterised by μ and Σ .
- Common in science due to the *central limit theorem* (CLT).
- A general PDF $p(x)$ has a shape parameterised by non-linear combinations of μ and Σ (c.f. moments)

Gaussian PDF

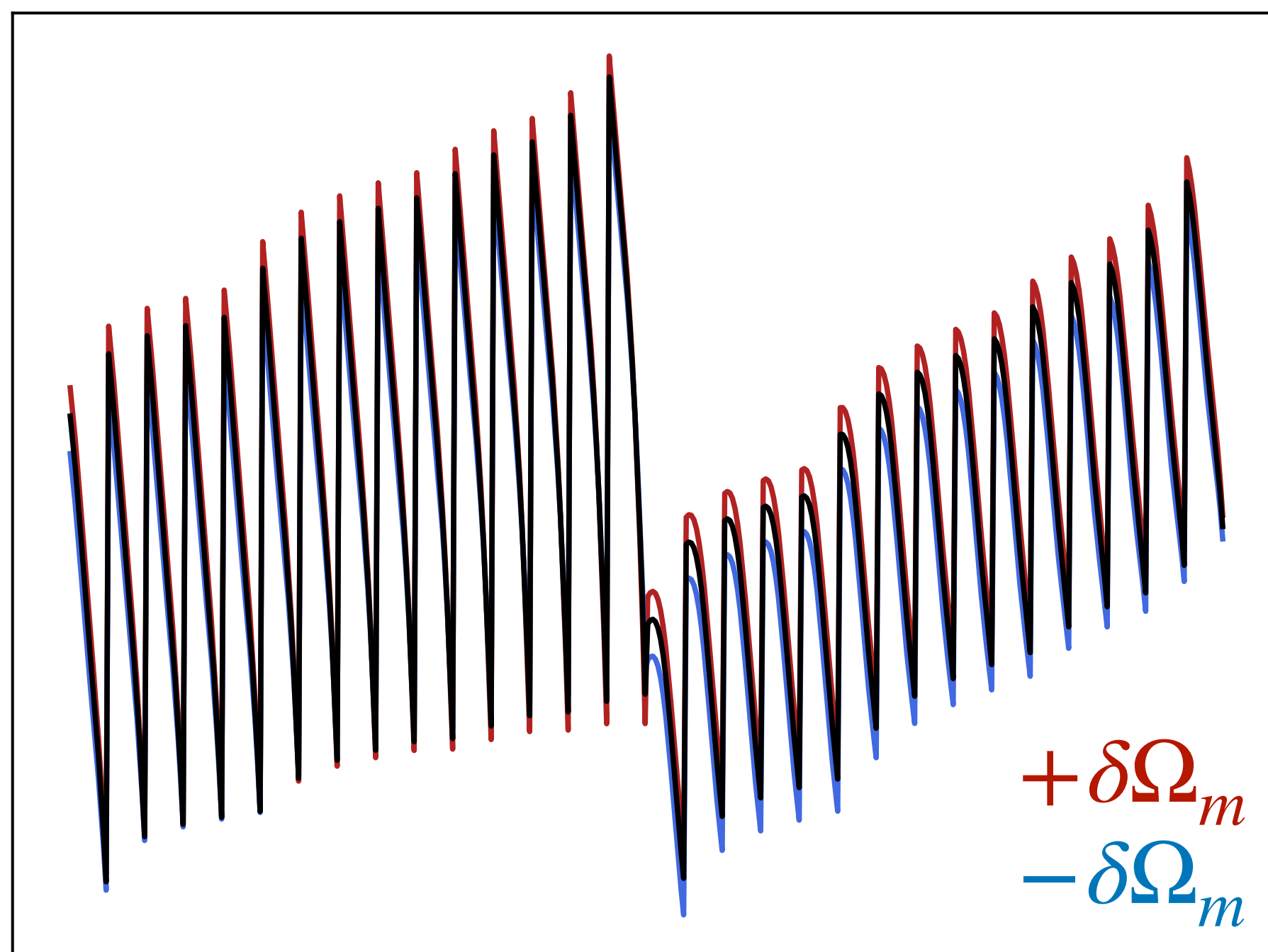
$$\mathcal{G}[\hat{x} | x[\pi], \Sigma] = \frac{1}{|2\pi\Sigma|} \exp \left[-\frac{1}{2} (\hat{x} - x[\pi])^T \Sigma^{-1} (\hat{x} - x[\pi]) \right]$$

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 - A
- A Gaussian PDF can be characterised by $\mu = x[\pi]$ and $\Sigma = \Sigma[\pi]$.
 - This corresponds to a physical model $x[\pi]$ corrupted into $\hat{x}$

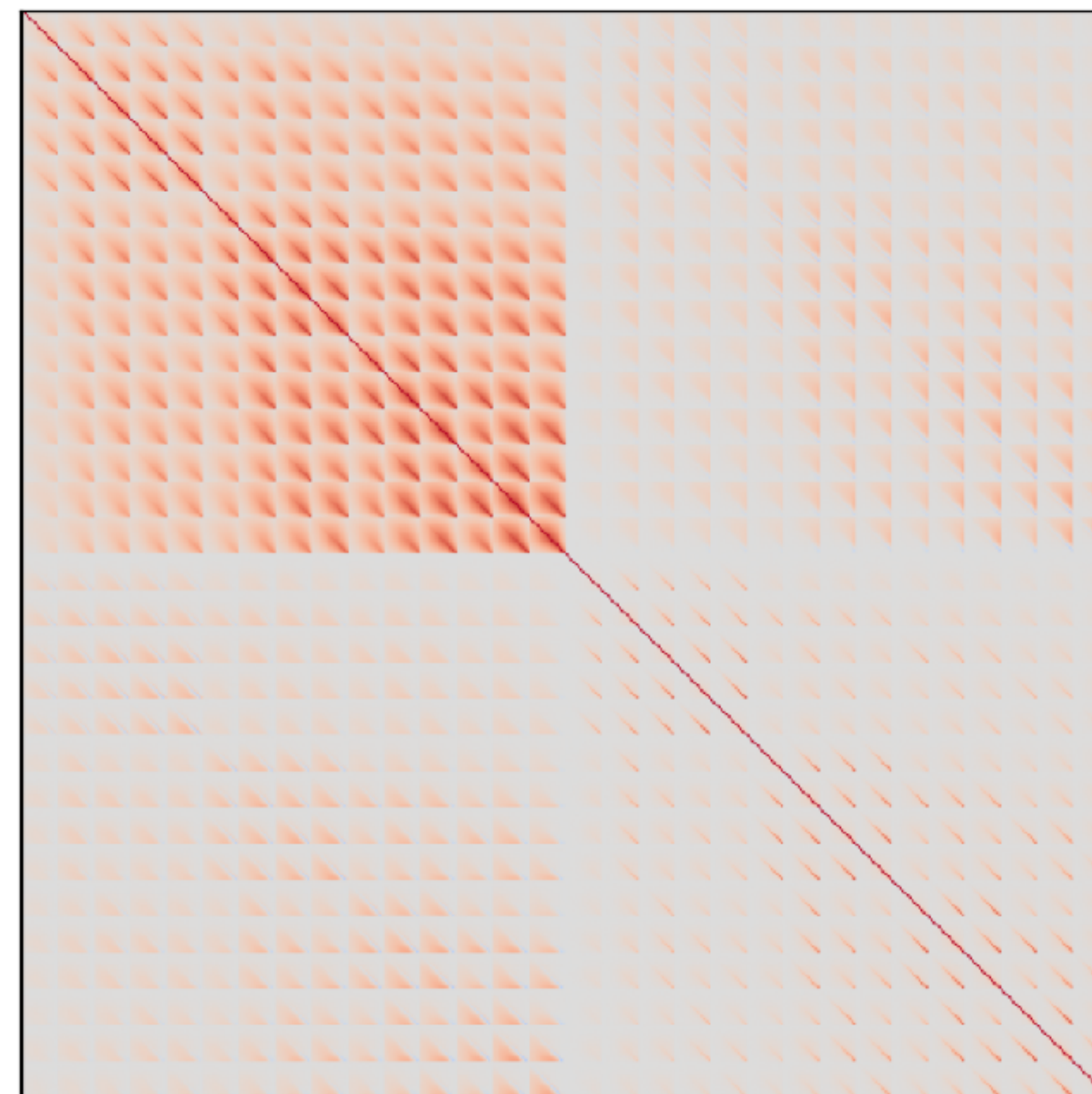
Σ (c.f. moments)

$$p(\hat{x} \mid \pi) = \mathcal{G} \left[\hat{x} \mid x[\pi], \Sigma \right]$$

$x[\pi]$

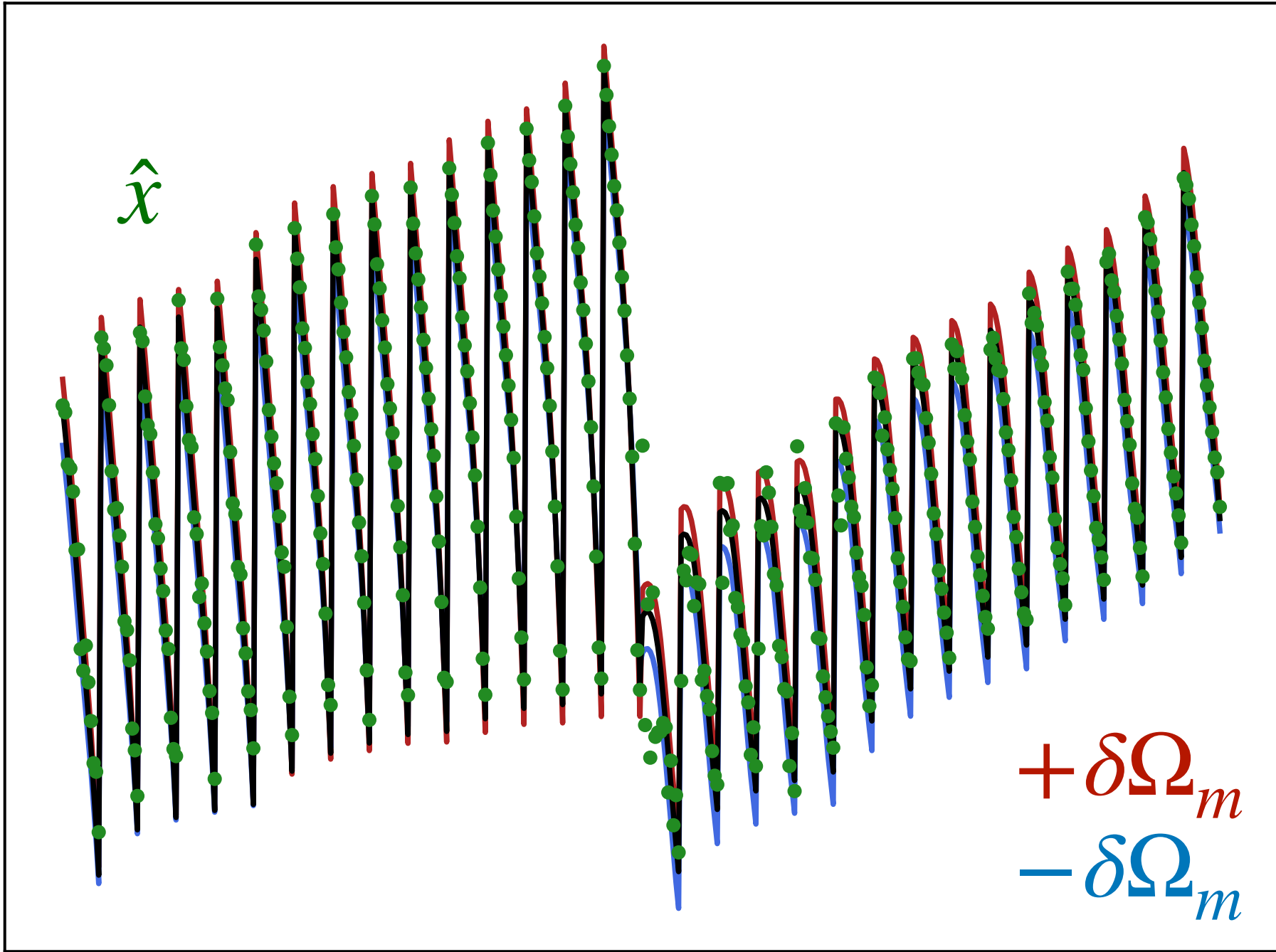


Σ

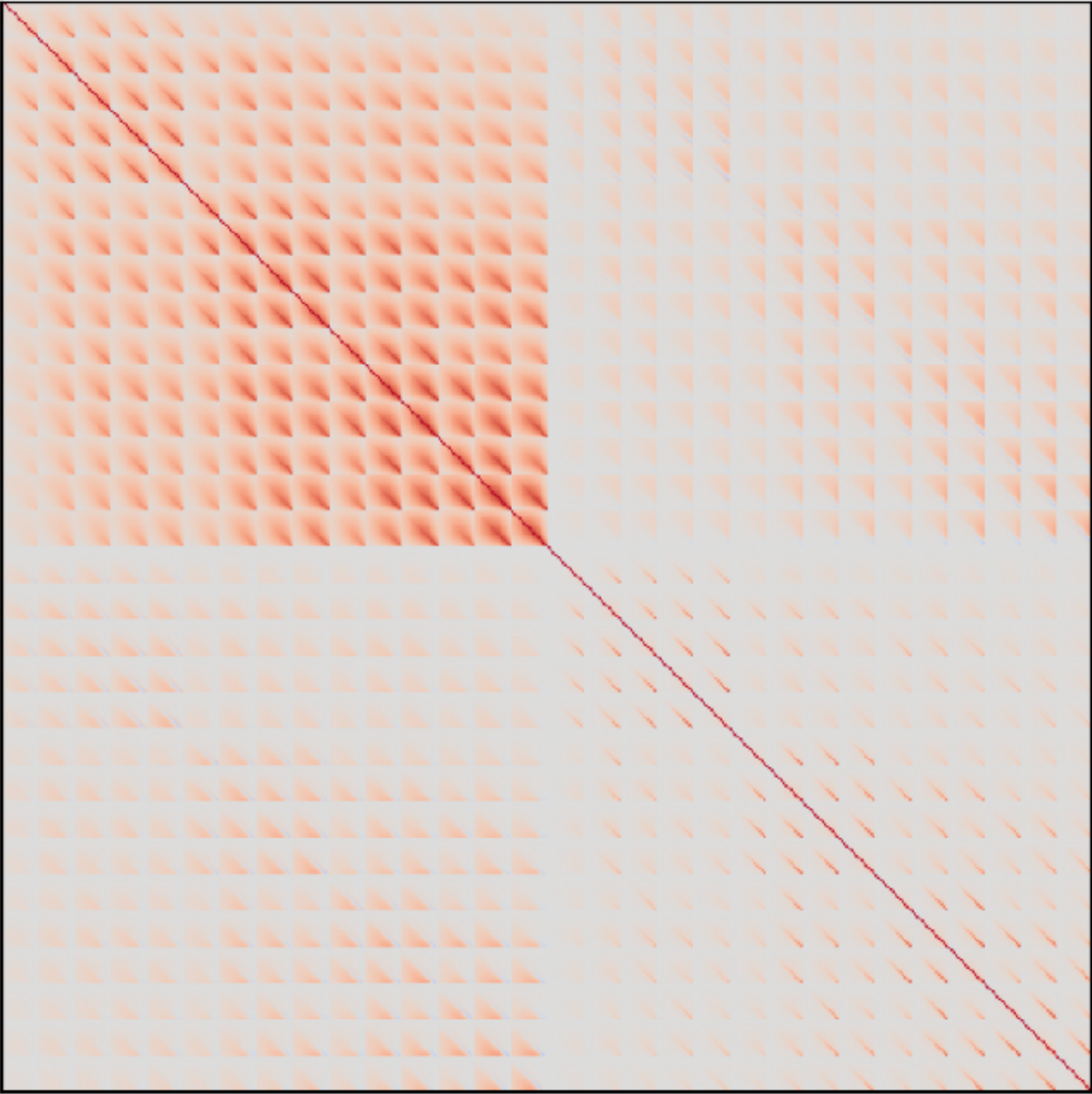


$$\hat{x} \sim \mathcal{G} \left[\hat{x} \mid x[\pi], \Sigma \right]$$

$x[\pi]$



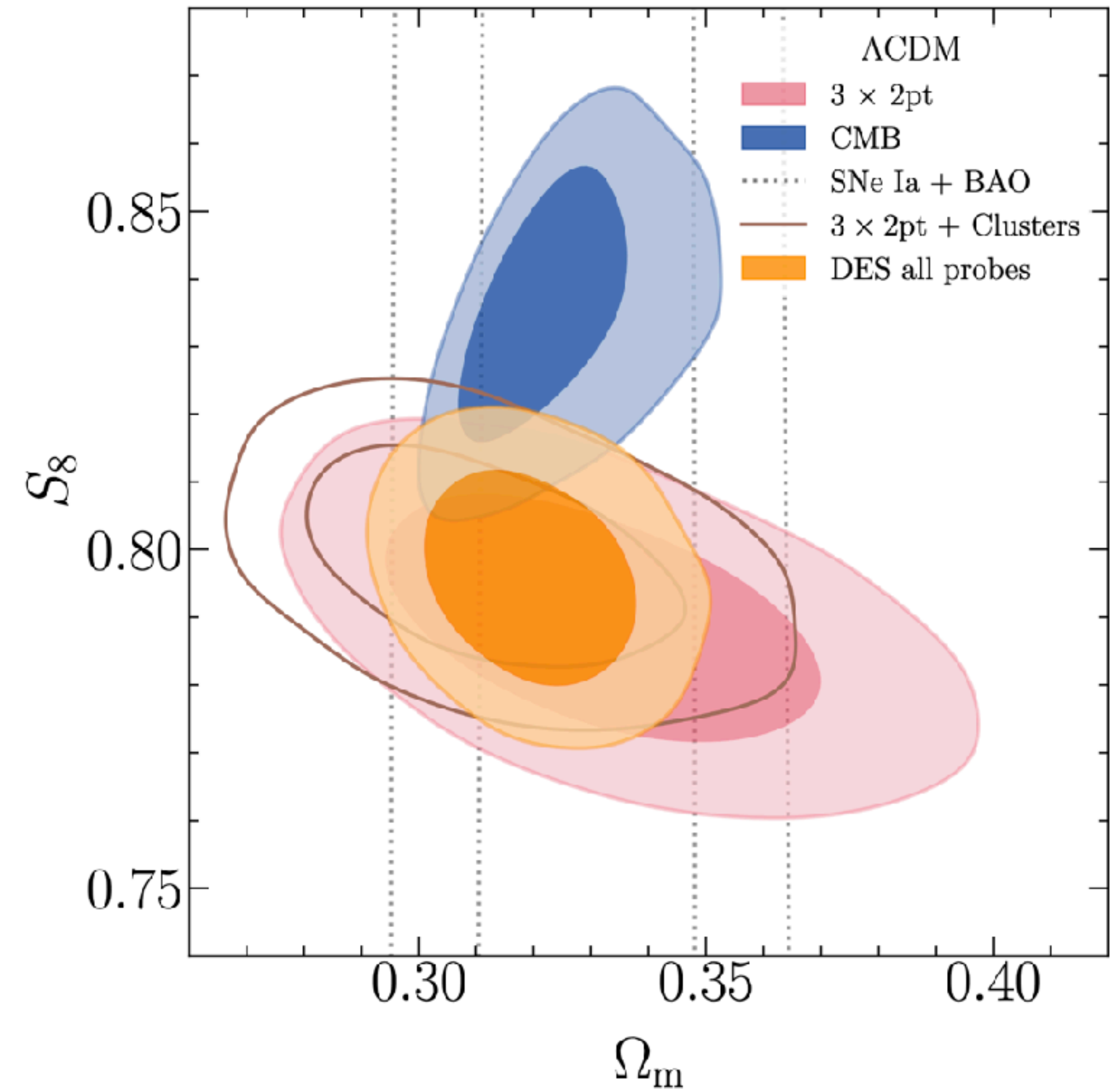
Σ



Bayesian Inference

- What does a Bayesian state as a result?

$$p(\pi | \hat{x}) = \frac{p(\hat{x} | \pi)p(\pi)}{p(\hat{x})}$$



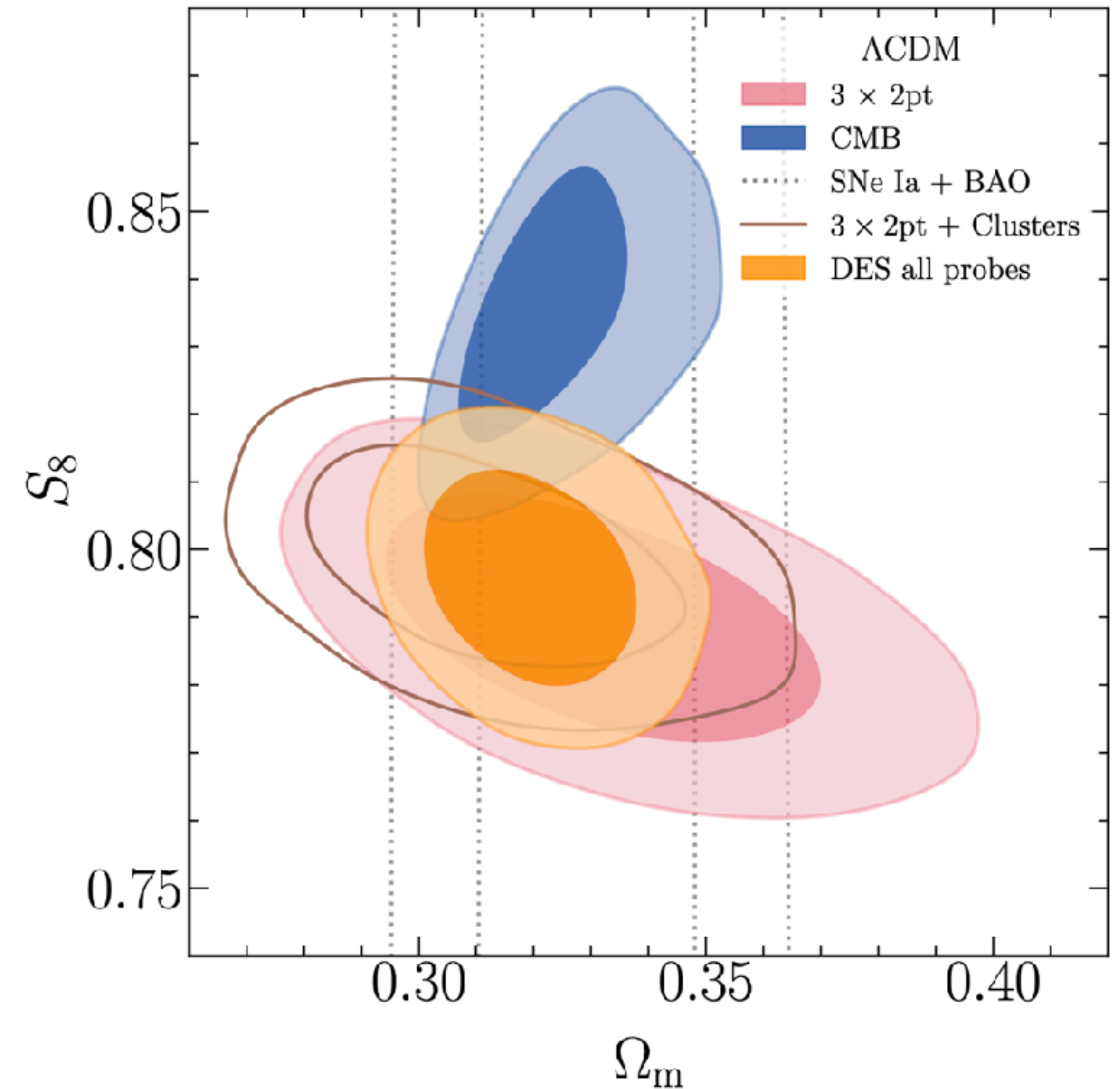
Dark Energy Survey 2026

Bayesian Inference

- What does a Bayesian state as a result?

$$p(\pi | \hat{x}) = \frac{p(\hat{x} | \pi)p(\pi)}{p(\hat{x})}$$

- How do you interpret this?



Dark Energy Survey 2026

Program of data analysis in physics

- Measure a datavector $\hat{x}$ that we assert is drawn from a PDF $p(x)$
- Build a theoretical model for this PDF, using some parameters π
 - We assert that these two PDFs are equal if we know the true π , i.e.
$$p(x | \pi) = p_{\text{model}}(x | \pi)$$
- We then want the values π that best fit the data $\hat{x}$

Program of data analysis in physics

- Measure a datavector $\hat{x}$ that we assert is drawn from a PDF $p(x)$

-

- What's wrong here?

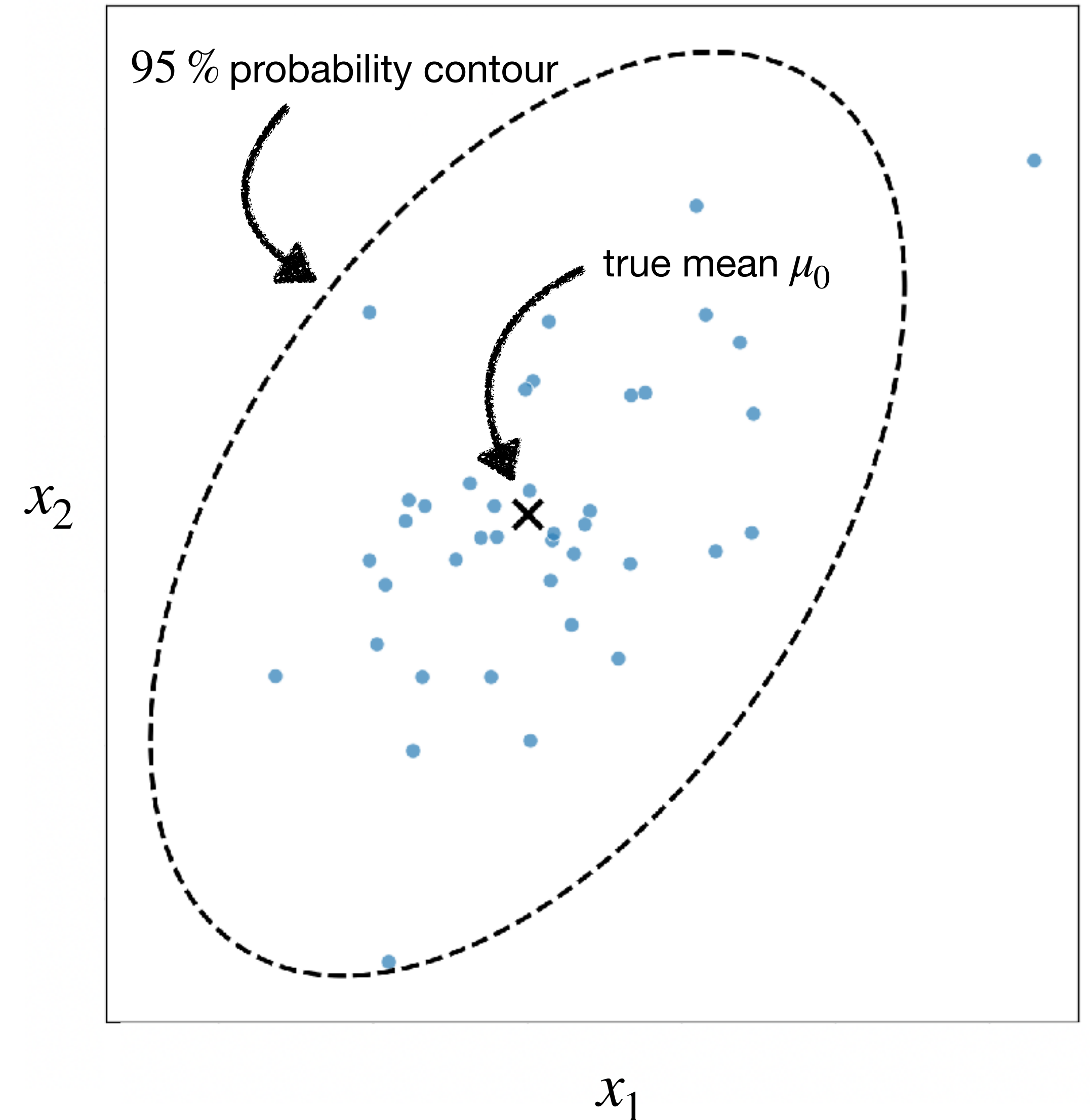
- Nature doesn't give us parameters, PDFs or estimators...

-

Setup

- We observe $\mathcal{X} = \{x_1, \dots, x_n\}$ where $x_i \sim \mathcal{G}[x | \mu, \Sigma]$.
- We know Σ .
- We want to infer μ from $\mathcal{X}$.

Samples $\mathcal{X} = \{x_1, \dots, x_n\} \sim \mathcal{G}(x | \mu, \Sigma)$



Problem

- Design an estimator $\hat{\mu}$ and calculate its sampling distribution $p(\hat{\mu} | \mu)$.
- We will use $\hat{\mu}_{\text{MLE}} = \bar{x}$,
- In this case, $p(\hat{\mu}_{\text{MLE}} | \mu) = \mathcal{G}[\hat{\mu}_{\text{MLE}} | \mu, n^{-1}\Sigma]$.

Bayesian

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- Set a PDF for the prior belief on μ before observing $\mathcal{X}$, then set a posterior:

$$p(\mu | \hat{\mu}) \propto p(\hat{\mu} | \mu)p(\mu)$$

Bayesian

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- Depends on the likelihood of the data $p(\hat{\mu} | \mu)$

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- Assume $p(\mu) = \mathcal{G}[\mu | \mu_p, \Sigma_p]$ so that $p(\mu | \hat{\mu})$ is analytic.

$$\hat{\mu}_{\text{MAP}} = \min_{\mu} p(\mu | \hat{\mu})$$

Bayesian

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Bayesian

- Assume $p(\mu) = \mathcal{G}[\mu \mid \mu_p, \Sigma_p]$ so that $p(\mu \mid \hat{\mu})$ is analytic.

$$\hat{\mu}_{\text{MAP}} = \max_{\mu} p(\mu \mid \hat{\mu})$$

- A Bayesian is interested in how much μ -space contains some fraction of the posterior probability.

$$\int_{\mu \in \Omega} d\mu p(\mu \mid \hat{\mu}) = 0.95$$

Practically speaking...

- Let $p(\mu) = \mathcal{G}[\mu \mid \mu_p, \Sigma_p]$. This is a *conjugate* prior on μ before any data.

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- We have $p(\hat{\mu} \mid \mu) = \mathcal{G}[\hat{\mu} \mid \mu, \Sigma/n]$.

- We can show that

$$p(\mu \mid \hat{\mu}) = \mathcal{G}[\mu \mid \mu_{\text{post}}, \Sigma_{\text{post}}]$$

- Where

$$\Sigma_{\text{post}} = \Sigma_p (\Sigma_p^{-1} + N\Sigma^{-1})^{-1}$$

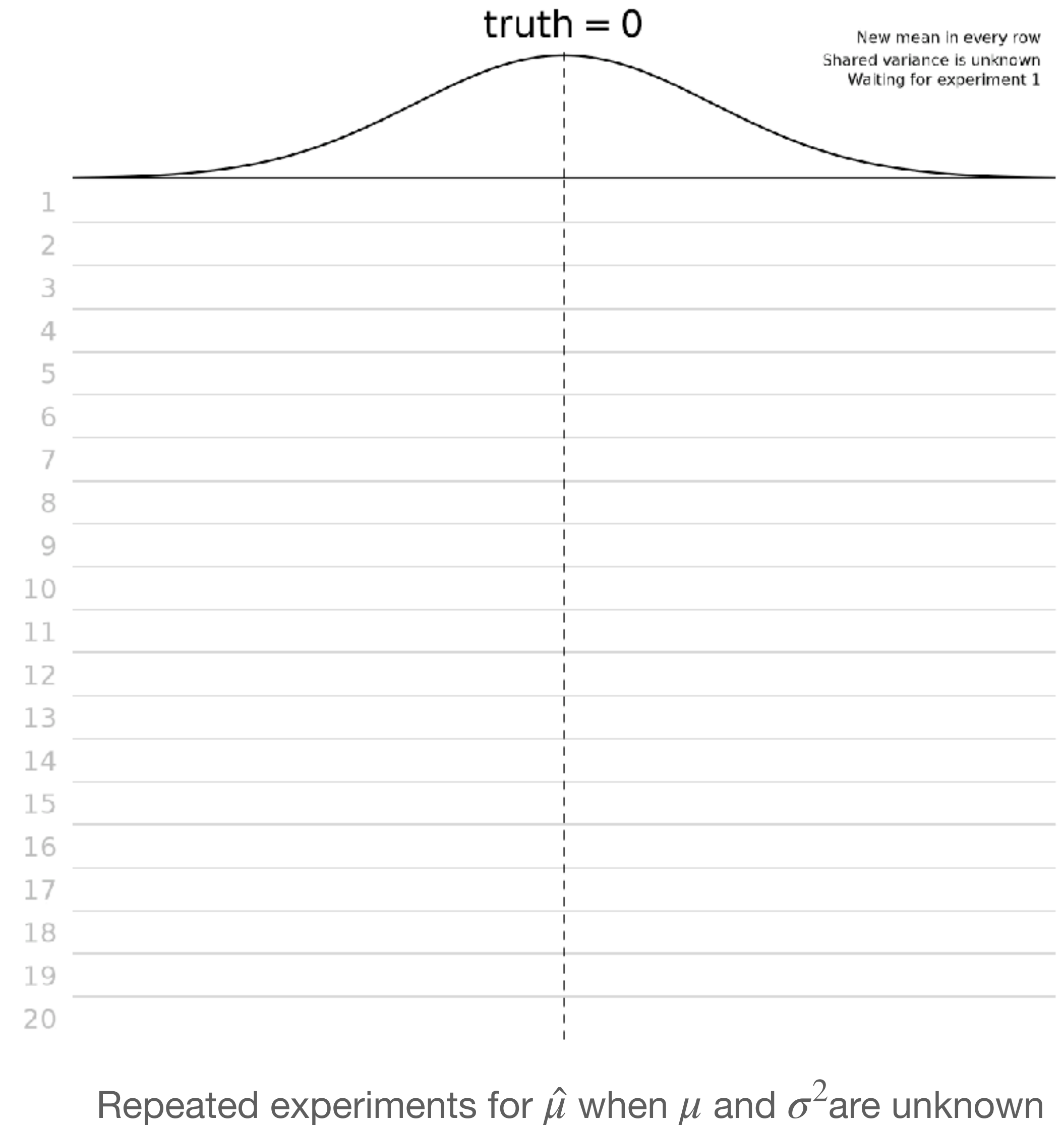
$$\mu_{\text{post}} = \Sigma_{\text{post}} (\Sigma_p^{-1} \mu_p + N\Sigma^{-1} \bar{x})$$

The Bayesian Credible Region (CR)

- What is the region of parameter space that encloses 95% of the posterior probability?
 - We'll demand that

$$\min_{\pi \in V_{95\%}} p(\pi | x) \geq \max_{\pi \notin V_{95\%}} p(\pi | x)$$

- This depends on the prior $p(\pi)$!



The Bayesian Credible Region (CR)

- In the Bayesian interpretation of probability:

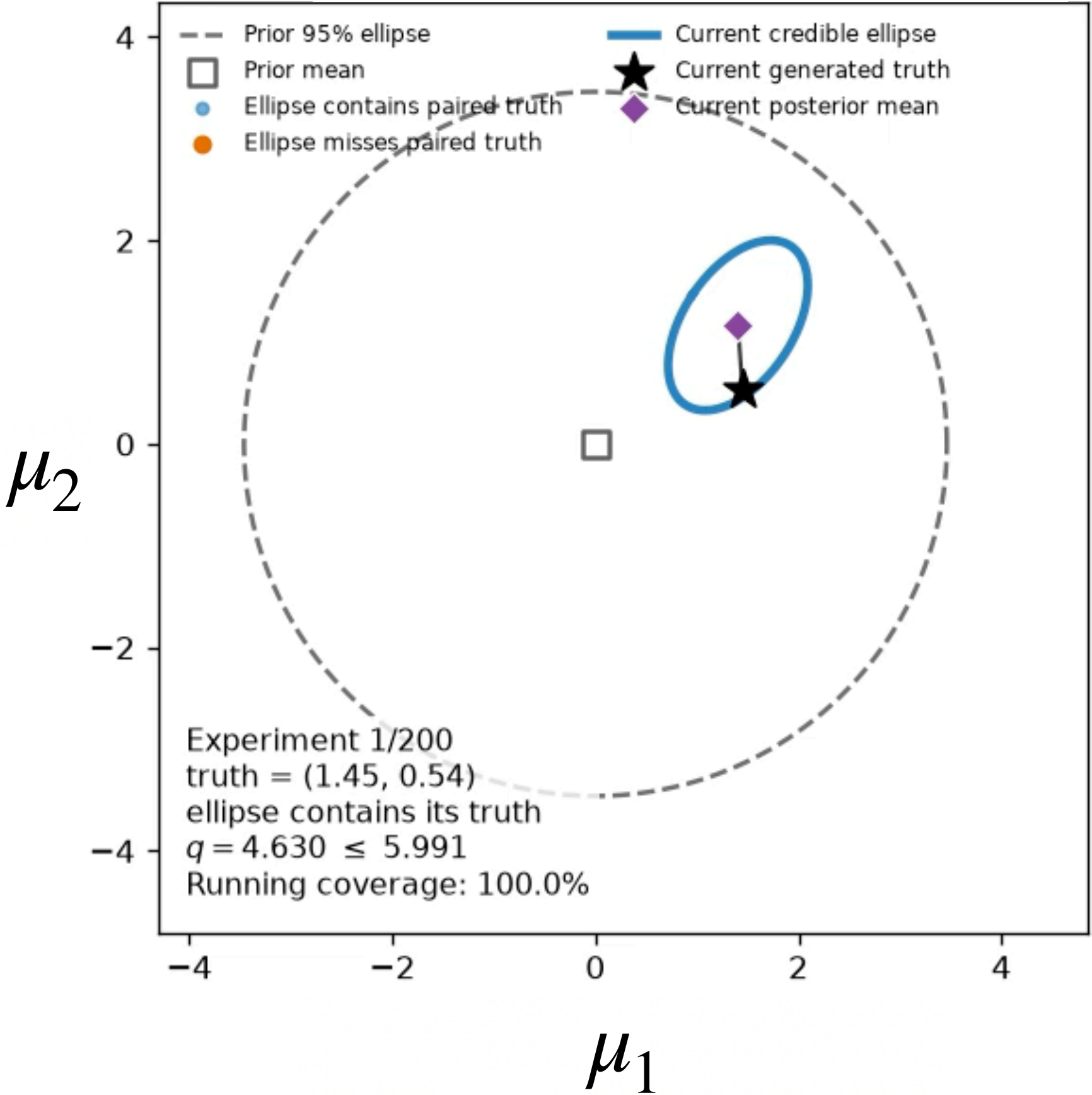
"Given the assumed prior $p(\mu)$ the interval

$$P \left[(\mu - \mu_{\text{post}})^{\top} \Sigma_{\text{post}}^{-1} (\mu - \mu_{\text{post}}) \leq \chi_{d, 0.95}^2 \mid \mathcal{X} \right] = 0.95.$$

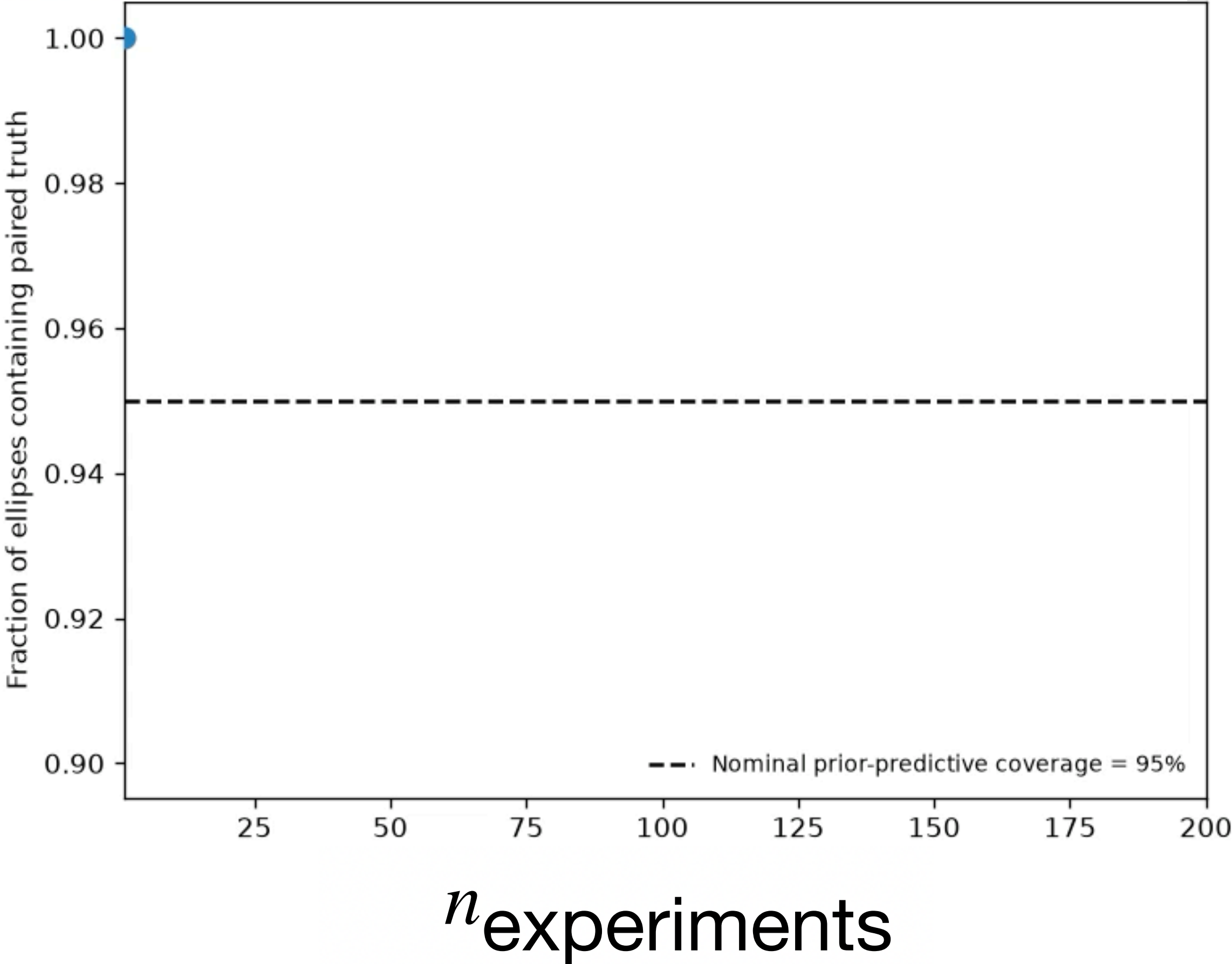
contains 95% of the posterior probability."

Example: The credible region $\mathcal{C}_{0.95}$ in repeated experiments

95% credible regions given repeated experiments



95% coverage fraction over independent experiments



The Bayesian analysis

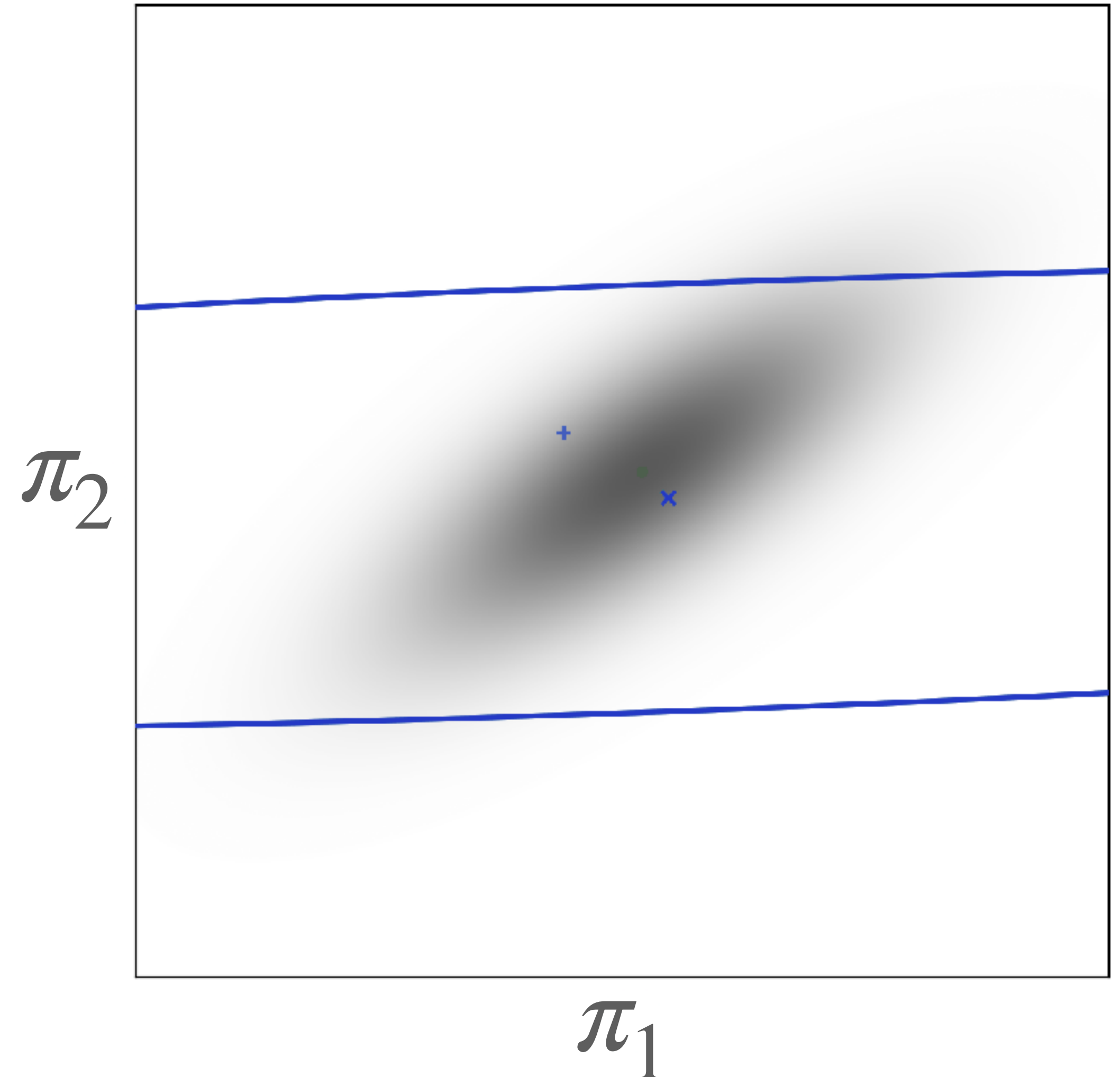
- The Bayesian measured a set of data $\mathcal{X}$, built a (perfectly accurate) model $\mathcal{G}[x | \mu, \Sigma]$
- They defined an estimator $\hat{\mu}$ which is unbiased and has minimum variance
 - The estimator had a known distribution $p(\hat{\mu} | \mu) = \mathcal{G}[\hat{\mu} | \mu, \Sigma/n]$
- They defined a prior PDF $\mu \sim p(\mu) = \mathcal{G}[\mu | \mu_p, \Sigma_p]$ which gave an analytic posterior $p(\mu | \Sigma)$
- They calculated a credible region $\mathcal{C}_{95\%}$: a region of μ -space that contains 95% of the posterior probability.

Limitations of the current problem

- We assumed Σ is known, for high-dimensional data, accurately estimating this matrix is the biggest problem in an analysis.
- Do you trust the result from this posterior?
- What happens in a parallel universe?
- What if $p(\pi | \hat{x}) \propto p(\hat{\pi} | \pi)p(\pi)$ is *not* analytic?

Priors and likelihood models

- Blue: wrong likelihood
- Pink: bad prior
- Mint: better prior



A bit more realism... $\mu = x[\pi]$ and $\Sigma = \hat{\Sigma}$

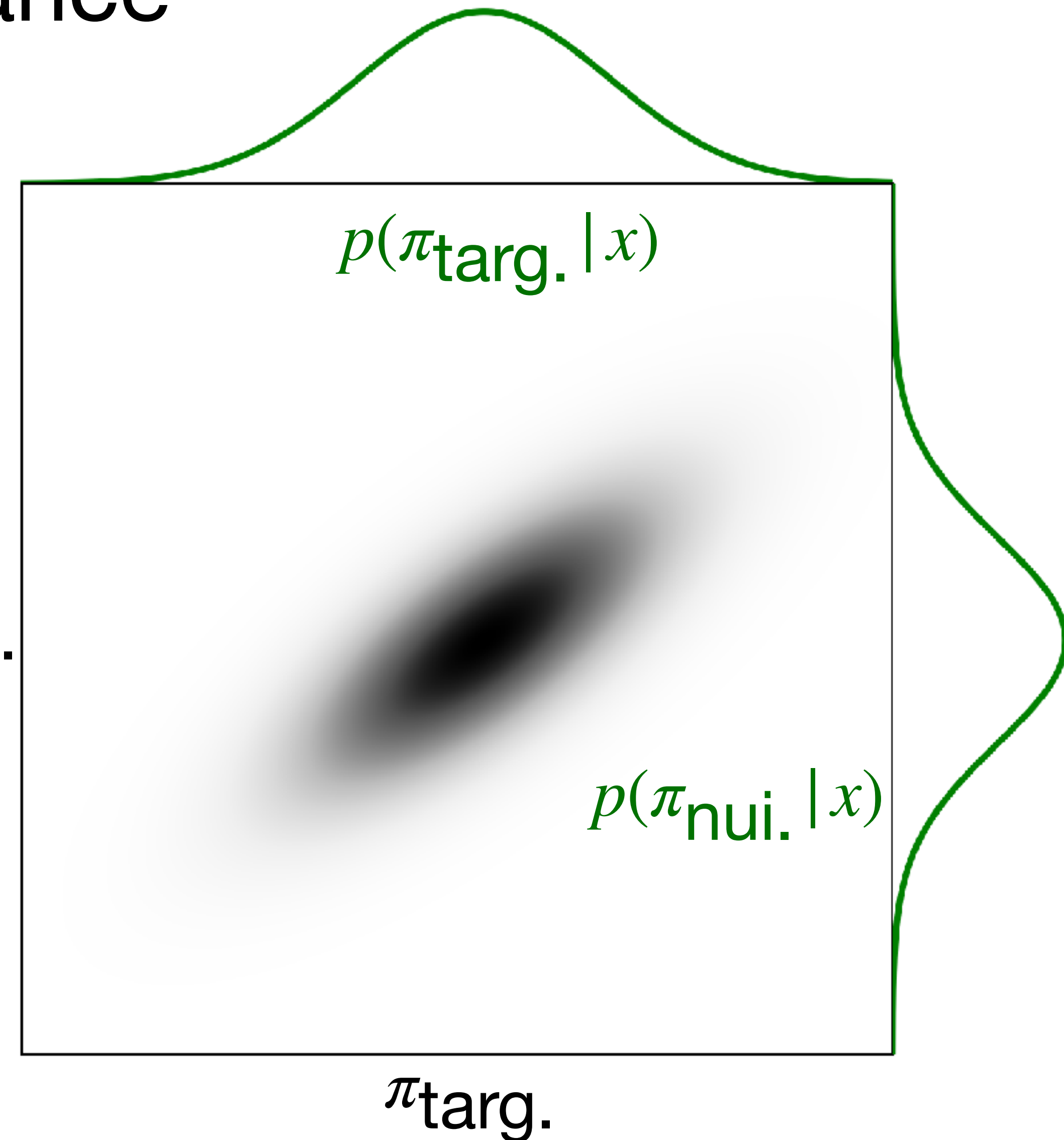
- We have assumed that we know the data covariance Σ .
- It is common to assume an expectation that is a function of some physical parameters π , e.g. $x \sim p(x | x[\pi], \pi)$
- Posterior $p(\pi | \hat{x}) \propto p(\hat{x} | \pi)p(\pi)$ may not be analytic: how do we sample from it to obtain a credible region?

Marginalisation over π_{nuisance}

- Marginalisation over posterior

$$p(\pi_{\text{targ.}} | x) = \int d\pi_{\text{nui.}} p(\pi_{\text{targ.}}, \pi_{\text{nui.}} | x)$$

$\pi_{\text{nui.}}$



Model comparison in Bayesian analyses

- The evidence Z given a model M normalises the posterior.
- It is calculated by averaging the likelihood over the prior

$$Z \equiv p(x | M) = \int d\pi p(x | \pi, M) p(\pi | M)$$

Model comparison in Bayesian analyses

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$$Z \equiv p(x | M) = \int d\pi p(x | \pi, M) p(\pi | M)$$

- This allows for parameter inference

$$p(\pi | x, M) = \frac{p(x | \pi, M) p(\pi | M)}{Z}.$$

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- Bayes allows us to assign odds to competing models, e.g. M_C v.s. M_I

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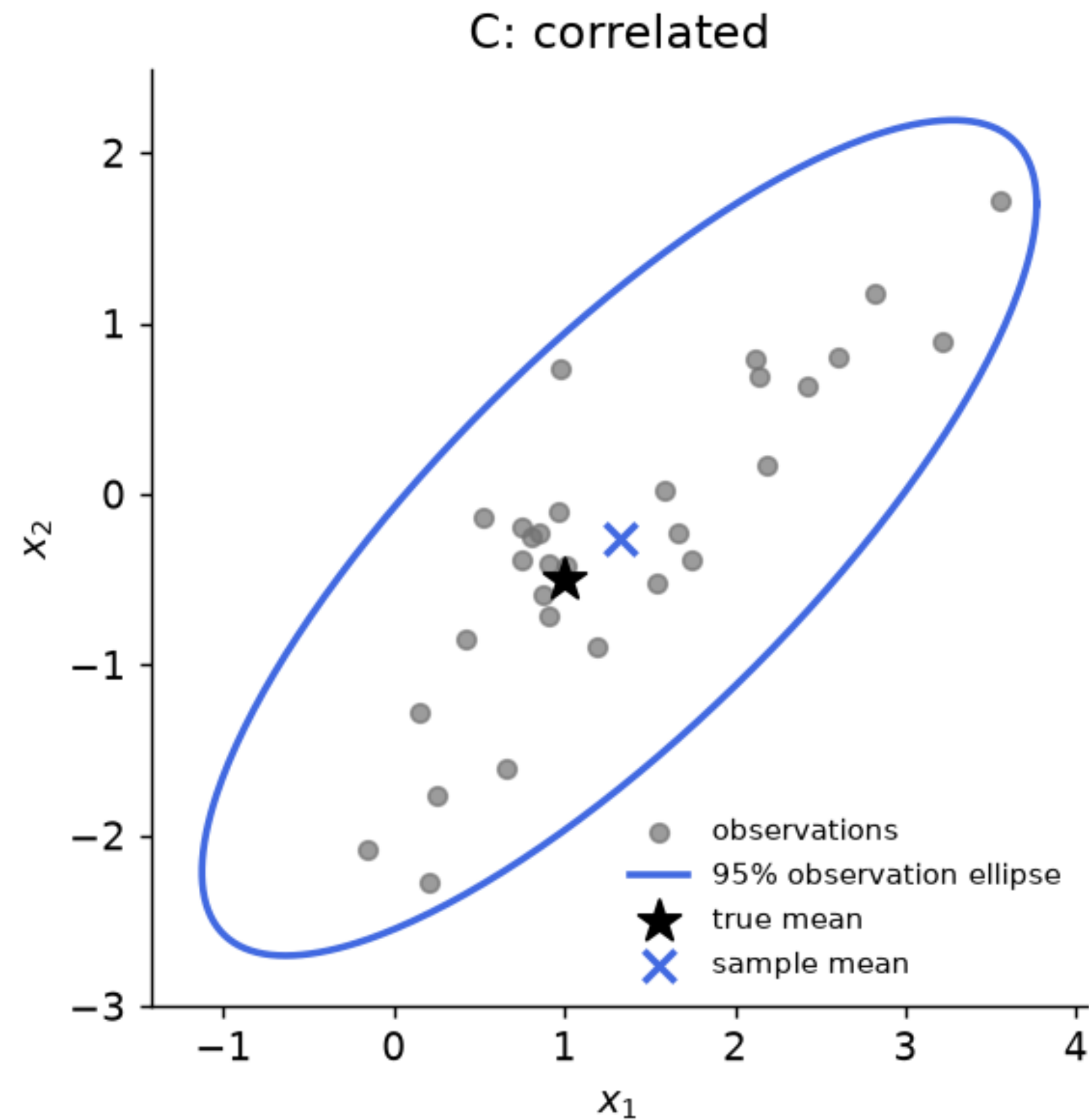
- A model M may have different parameters, a different prior, a different likelihood, ...

But today we keep it simple with only a different covariance Σ

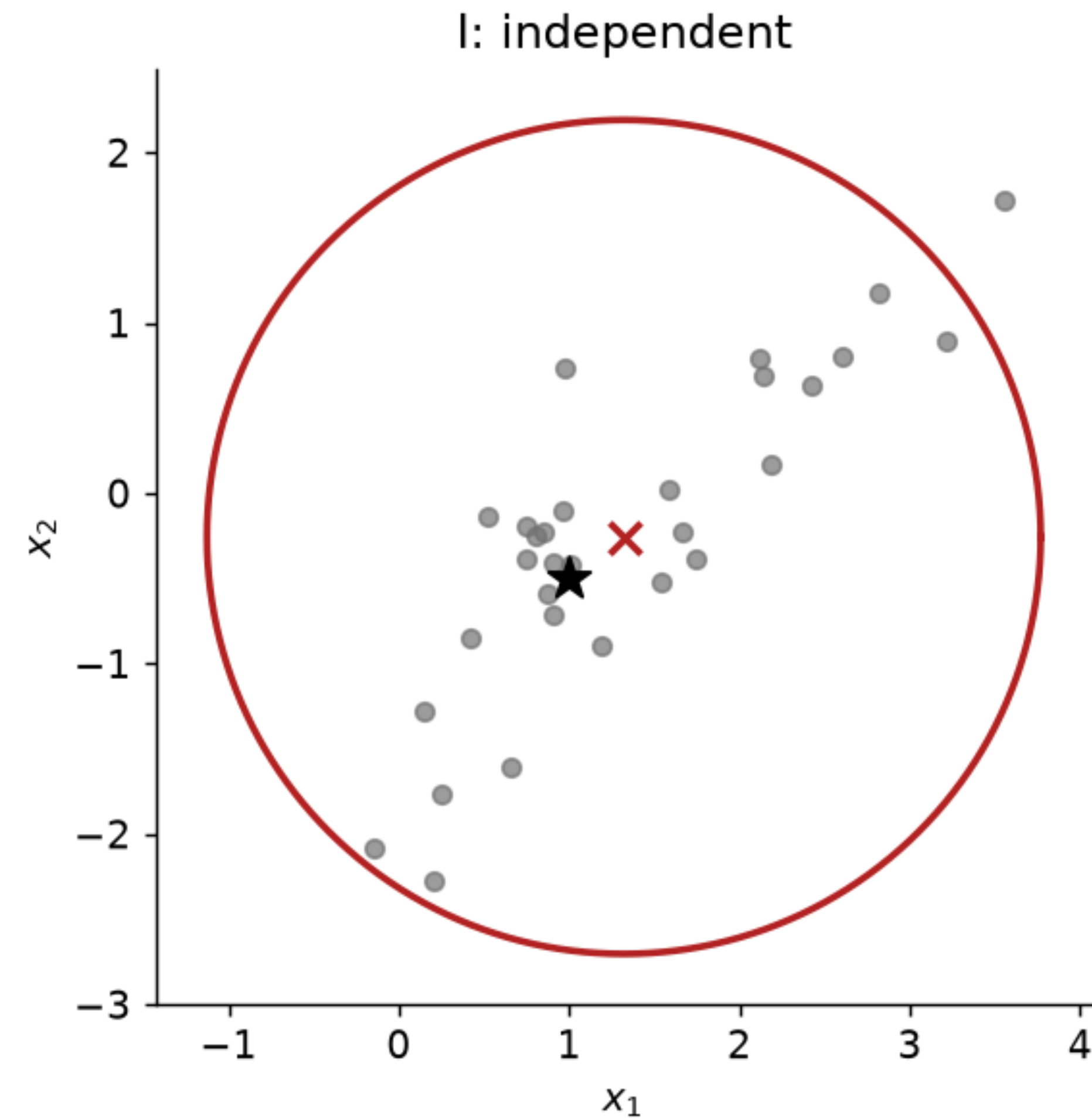
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Model comparison in Bayesian analyses



M_C



M_I

Model comparison in Bayesian analyses

- The Bayes Factor B_{CI} quantifies the odds between two models

$$\frac{P(M_C | D)}{P(M_I | D)} = \underbrace{\frac{Z_C}{Z_I}}_{B_{CI}} \frac{P(M_C)}{P(M_I)}.$$

- $B_{CI} > 1$ favours M_C over M_I ,
- $\log B_{CI} = 0$ means data did not prefer either, but can still constrain e.g. μ

Model comparison in Bayesian analyses

- Evidence Z supports relative comparisons between models, but it is not an absolute measure of whether a model is adequate or not.

$$\frac{P(M_C | D)}{P(M_I | D)} = \underbrace{\frac{Z_C}{Z_I}}_{B_{CI}} \frac{P(M_C)}{P(M_I)}.$$

Model comparison in Bayesian analyses

- Evidence Z supports relative comparisons between models, but it is not an absolute measure of whether a model is adequate or not.

$$\log Z = -\frac{nd}{2} \log(2\pi) - \frac{n-1}{2} \log |\Sigma|$$

$$W = C_0 + \Sigma/n$$

$$\delta = \bar{\mathbf{x}} - \mathbf{m}_0$$

$$-\frac{d}{2} \log n - \frac{1}{2} \text{tr}(\Sigma^{-1}S) - \frac{1}{2} \log |W| - \frac{1}{2} \delta^\top W^{-1} \delta.$$

- This analytic result depends on our simple data model, it is usually prohibitively expensive to calculate in higher dimensions.

Model comparison in Bayesian analyses

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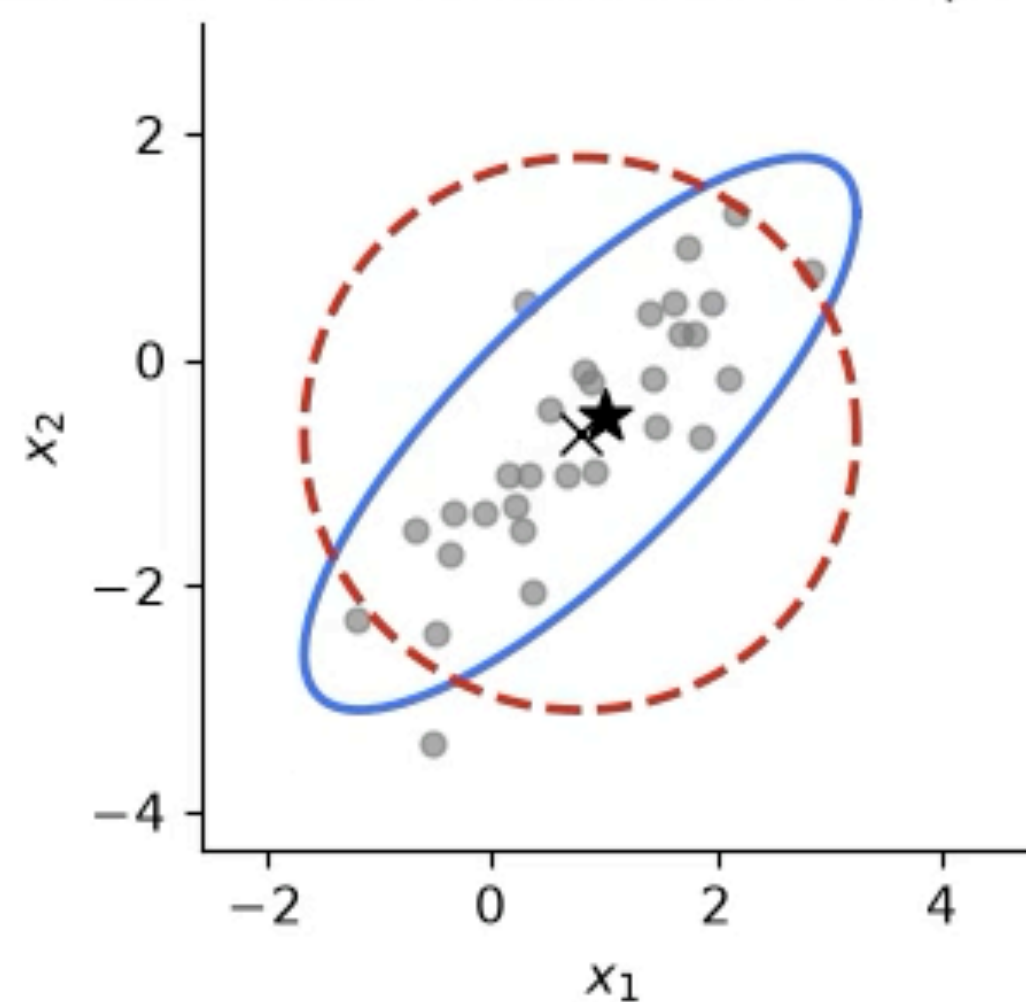
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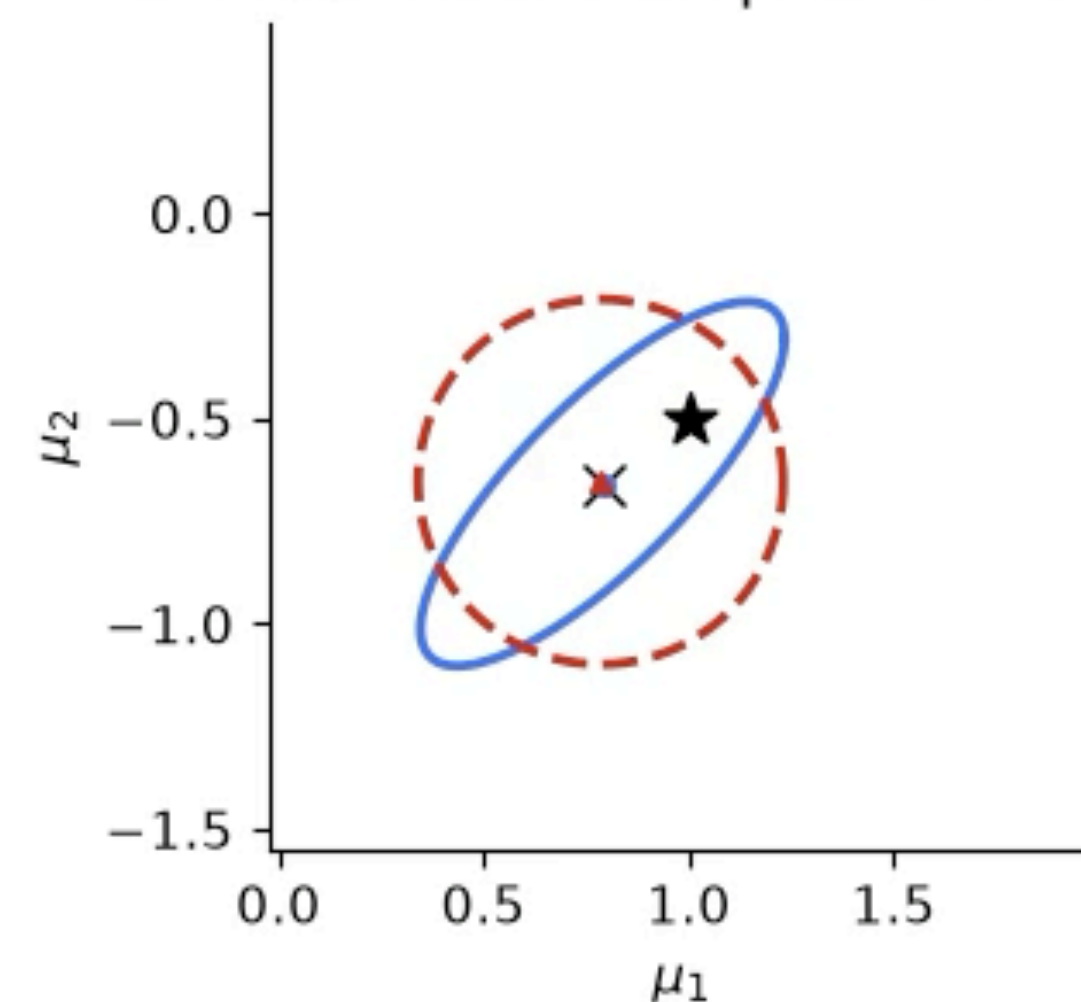
- This analysis is prohibitive

- Calculate Z_C, Z_I for differing covariances, then $B_{CI} = \frac{Z_C}{Z_I}$ to compare models

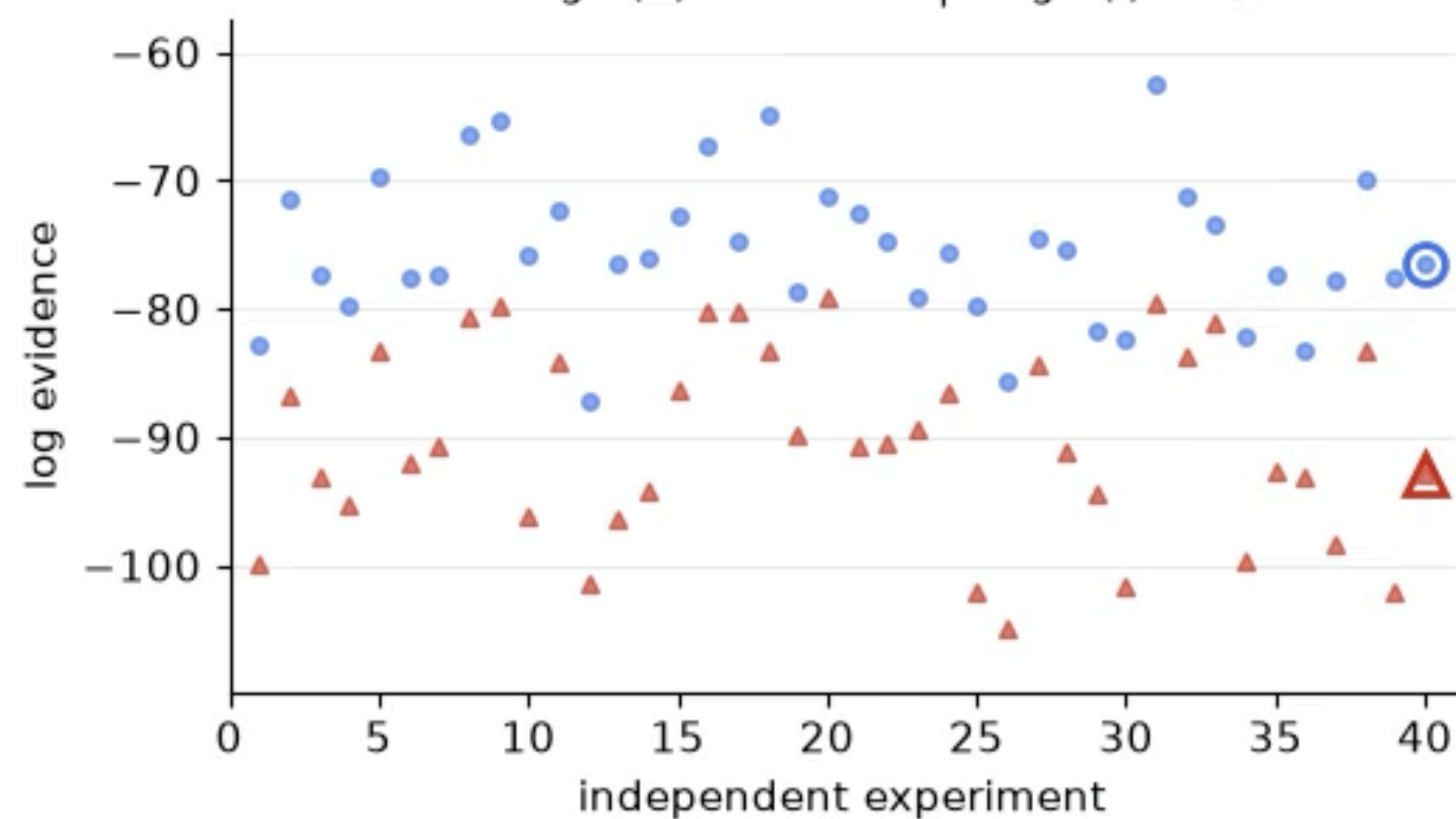
Data and 95% observation contours (at fitted mean)



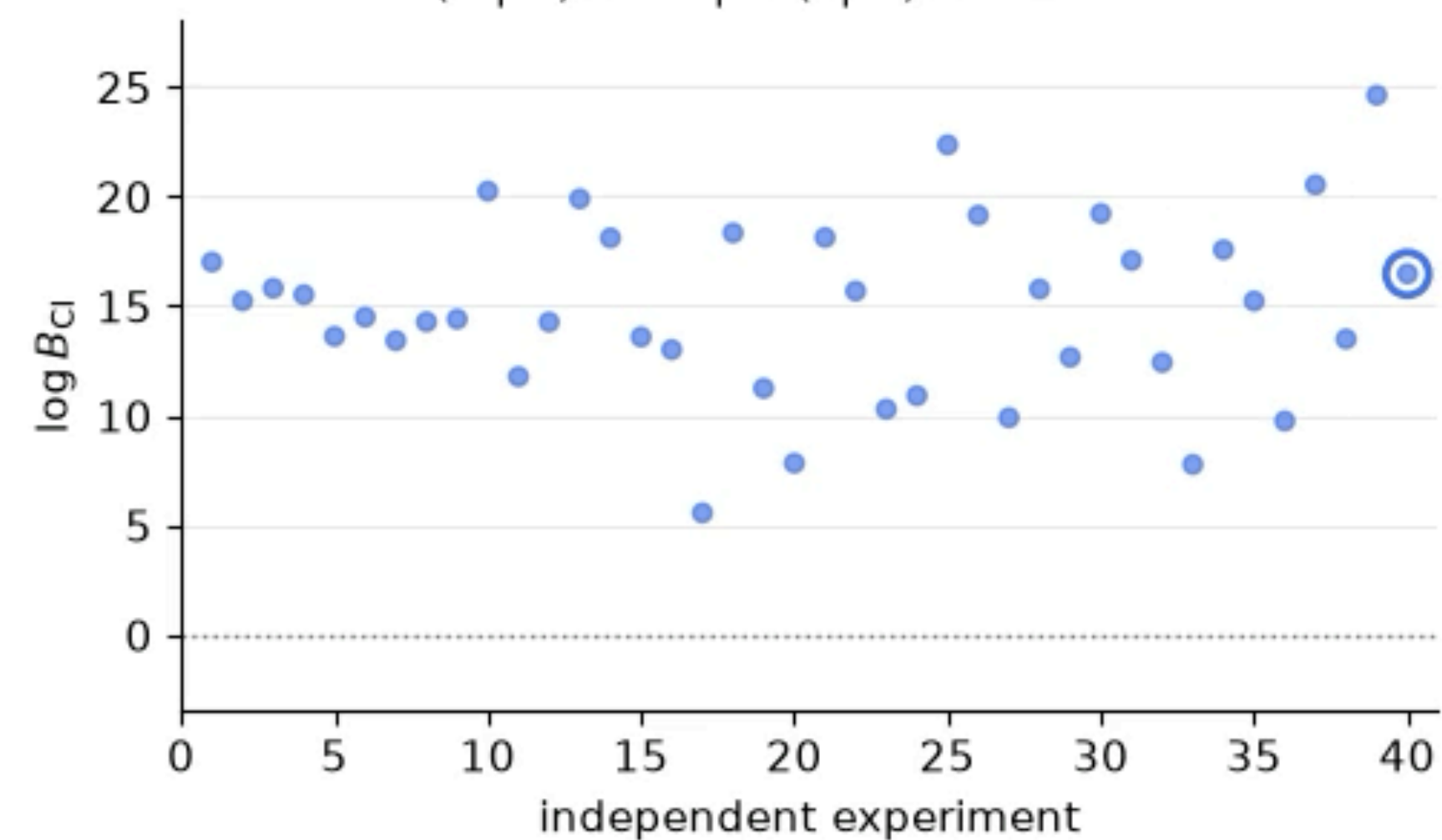
Mean inference: 95% posterior contours



Log evidence for each experiment
Current: $\log Z(C) = -76.47$ | $\log Z(I) = -92.93$



Current: $\log B(C/I) = +16.46$
 $P(C | D) \approx 1$ | $P(I | D) \approx 7.075e-08$



—●— C: correlated —▲— I: independent ● observations ★ true mean × sample mean

Write down something you learned today and
something you need to hear more about next lecture

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